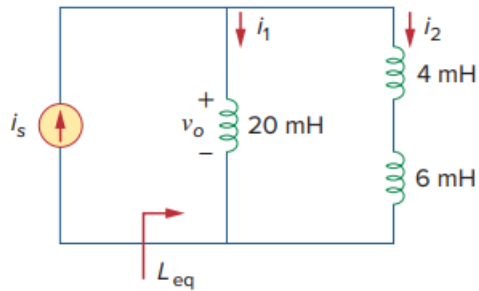


Problem

Consider the circuit in Fig. 6.83. Find: (a) L_{eq} , $i_1(t)$, and $i_2(t)$ if $i_s = 3e^{-t}$ mA, (b) $v_o(t)$, (c) energy stored in the 20-mH inductor at $t = 1$ s.

Fig. 6.83:



Step-by-step solution

Step 1 of 5

Refer to Figure 6.83 in the textbook.

(a)

Calculate the equivalent inductance seen from the source.

$$\begin{aligned} L_{eq} &= (4 \times 10^{-3} + 6 \times 10^{-3}) \parallel (20 \times 10^{-3}) \\ &= (10 \times 10^{-3}) \parallel (20 \times 10^{-3}) \\ &= \frac{(10 \times 10^{-3})(20 \times 10^{-3})}{(10 \times 10^{-3}) + (20 \times 10^{-3})} \\ &= 6.667 \text{ mH} \end{aligned}$$

Therefore, the equivalent inductance, L_{eq} is **6.667 mH**.

Step 2 of 5

The source current is,

$$i_s = 3e^{-t} \text{ mA}$$

From Figure 6.83 in the textbook, calculate the current through 20 mH inductor using current division rule.

$$\begin{aligned} i_1(t) &= i_s \left(\frac{4 \times 10^{-3} + 6 \times 10^{-3}}{4 \times 10^{-3} + 6 \times 10^{-3} + 20 \times 10^{-3}} \right) \\ &= (3 \times 10^{-3} e^{-t}) \left(\frac{10}{30} \right) \\ &= 1e^{-t} \text{ mA} \end{aligned}$$

Therefore, the current, $i_1(t)$ is **$1e^{-t}$ mA**.

Step 3 of 5

From Figure 6.83 in the textbook, calculate the current through 4 mH inductor using current division rule.

$$\begin{aligned} i_2(t) &= i_s \left(\frac{20 \times 10^{-3}}{4 \times 10^{-3} + 6 \times 10^{-3} + 20 \times 10^{-3}} \right) \\ &= \left(3 \times 10^{-3} e^{-t} \right) \left(\frac{20}{30} \right) \\ &= 2e^{-t} \text{ mA} \end{aligned}$$

Therefore, the current, $i_2(t)$ is $2e^{-t} \text{ mA}$.

Step 4 of 5

(b)

Calculate the voltage across the 20 mH inductor.

$$v_o(t) = L \frac{di_1(t)}{dt}$$

Substitute 20 mH for L and $1e^{-t} \text{ mA}$ for $i_1(t)$.

$$\begin{aligned} v_o(t) &= \left(20 \times 10^{-3} \right) \frac{d(1 \times 10^{-3} e^{-t})}{dt} \\ &= \left(20 \times 10^{-6} \right) \frac{d(e^{-t})}{dt} \\ &= \left(20 \times 10^{-6} \right) e^{-t} (-1) \\ &= -20e^{-t} \mu\text{V} \end{aligned}$$

Therefore, the voltage across the 20 mH inductor, $v_o(t)$ is $-20e^{-t} \mu\text{V}$.

Step 5 of 5

(c)

Calculate the energy stored in the 20 mH inductor.

$$w(t) = \frac{1}{2} L i_1^2$$

Substitute 20 mH for L and $1e^{-t} \text{ mA}$ for $i_1(t)$.

$$\begin{aligned} w(t) &= \frac{1}{2} \left(20 \times 10^{-3} \right) \left(1 \times 10^{-3} e^{-t} \right)^2 \\ &= 10 \times 10^{-9} e^{-2t} \text{ J} \end{aligned}$$

Calculate the energy stored in the 20 mH inductor at $t = 1 \text{ s}$.

$$\begin{aligned} w(1) &= 10 \times 10^{-9} e^{-2(1)} \\ &= 10 \times 10^{-9} e^{-2} \\ &= 1.35 \times 10^{-9} \text{ J} \\ &= 1.35 \text{ nJ} \end{aligned}$$

Therefore, the energy stored in the 20 mH inductor at $t = 1 \text{ s}$ is 1.35 nJ .